Light

GCSE Computer Science — CPU Architecture (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A smartphone runs multiple apps simultaneously while playing music in the background.
Which component is responsible for processing all these tasks?

- ○ **A.** The Central Processing Unit
- ○ **B.** The display screen
- ○ **C.** The battery
- ○ **D.** The speaker

Q2 (1 mark)

A security system checks if BOTH a valid key card AND a correct PIN have been entered.
Which CPU component evaluates this AND condition?

- ○ **A.** The Memory Data Register
- ○ **B.** The Arithmetic Logic Unit
- ○ **C.** The cache memory
- ○ **D.** The Program Counter

Q3 (1 mark)

Different CPU components must work together in a specific sequence. Which component ensures all parts operate at the correct time?

- ○ **A.** The Accumulator
- ○ **B.** The cache memory
- ○ **C.** The Control Unit
- ○ **D.** The hard drive

Q4 (1 mark)

When a computer is switched off, what happens to data stored in the CPU's cache?

- ○ **A.** It is lost because cache is volatile memory
- ○ **B.** It is automatically saved to the hard drive
- ○ **C.** It remains stored permanently
- ○ **D.** It is transferred to a USB drive

Q5 (1 mark)

Compared to RAM, how do CPU registers differ in quantity?

- ○ **A.** There are billions of registers in a typical CPU
- ○ **B.** There are only a small number of registers, typically fewer than 100
- ○ **C.** Registers and RAM have the same capacity
- ○ **D.** There are no registers in modern CPUs

Q6 (1 mark)

In Von Neumann architecture, instructions and data travel to the CPU using:

- ○ **A.** Completely separate pathways that never interact
- ○ **B.** Wireless signals between components
- ○ **C.** The same shared bus system
- ○ **D.** Different coloured wires for each type

Q7 (1 mark)

A computer has a 32-bit Memory Address Register (MAR).
What does this 32-bit size determine?

- ○ **A.** The speed of the processor
- ○ **B.** The number of cores in the CPU

- ○ **C.** The size of the hard drive
- ○ **D.** The maximum amount of RAM that can be addressed

Q8 (1 mark)

A CPU has a 64-bit Memory Data Register (MDR).
What does this specification indicate?

- ○ **A.** The CPU can transfer 64 bits of data to or from memory in one operation
- ○ **B.** The CPU has 64 cores
- ○ **C.** The CPU runs at 64 GHz
- ○ **D.** The CPU has 64 MB of cache

Q9 (1 mark)

A program contains a loop that repeats 10 times.
How does the Program Counter behave during this loop?

- ○ **A.** The PC stops incrementing until the loop finishes
- ○ **B.** The PC is set to a previous address to repeat the loop instructions
- ○ **C.** The PC counts from 1 to 10
- ○ **D.** The PC is not used during loops

Q10 (1 mark)

A program calculates: result = (5 + 3) × 2

How does the Accumulator help with this chained calculation?

- ○ **A.** The Accumulator performs both operations simultaneously
- ○ **B.** The Accumulator is not used for mathematical operations
- ○ **C.** The Accumulator stores the intermediate result (8) so it can be used in the next calculation
- ○ **D.** The Accumulator sends the calculation to the hard drive

Q11 (1 mark)

A computer's CPU is upgraded from 2.4 GHz to 3.6 GHz.
What change does this represent?

- ○ **A.** A 50% increase in clock cycles per second
- ○ **B.** A 50% increase in the number of CPU cores
- ○ **C.** A 50% increase in RAM capacity
- ○ **D.** A 50% increase in storage space

Q12 (1 mark)

A quad-core CPU is running four different applications.
What advantage does having four cores provide?

- ○ **A.** Each application runs four times faster
- ○ **B.** Each application can potentially run on its own core simultaneously
- ○ **C.** Each core has four times more cache

○ **D.** Each application uses four times more power

Q13 (2 marks)

A computer system uses three types of buses: address bus, data bus, and control bus.
Which statement correctly describes the function of each bus?

○ **A.** Address bus carries data, data bus carries addresses, control bus carries programs

○ **B.** All three buses carry the same type of information but at different speeds

○ **C.** Address bus connects to the monitor, data bus to the keyboard, control bus to the mouse

○ **D.** Address bus carries memory locations, data bus carries values being transferred, control bus carries timing and command signals

Q14 (2 marks)

During the execute stage of the fetch-decode-execute cycle, which actions might occur?

○ **A.** The ALU performs a calculation, data is transferred between registers, or a result is written to memory

○ **B.** The instruction is retrieved from memory and placed in the MDR

○ **C.** The Control Unit interprets the instruction's opcode

○ **D.** The Program Counter is loaded with the first instruction's address

Q15 (2 marks)

When the CPU needs data, it searches the cache levels in a specific order.
What is this search order and why?

- ○ **A.** L3 first, then L2, then L1 — because L3 is largest and most likely to have the data
- ○ **B.** All levels are searched simultaneously to save time
- ○ **C.** L1 first, then L2, then L3 — because L1 is fastest, so checking it first minimises access time
- ○ **D.** The order is random and varies with each access

Q16 (2 marks)

The Current Instruction Register (CIR) holds the instruction currently being executed.
Why is this register necessary?

- ○ **A.** To count how many instructions have been completed
- ○ **B.** To keep the instruction available while the Control Unit decodes it and the CPU executes it
- ○ **C.** To store all instructions in the program
- ○ **D.** To send instructions directly to the monitor

Q17 (2 marks)

The CPU's internal clock generates a regular pulse signal.
What is the purpose of this clock signal?

- ○ **A.** To display the current time on the screen

- ○ **B.** To measure how long programs take to run
- ○ **C.** To power the CPU's transistors
- ○ **D.** To synchronise operations so all CPU components work in step with each other

Q18 (2 marks)

An 8-bit register is used to store the result of adding 200 + 100.
What problem occurs and what is it called?

- ○ **A.** An overflow error occurs because 300 exceeds the maximum value an 8-bit register can hold (255)
- ○ **B.** The result (300) is stored correctly with no issues
- ○ **C.** The calculation is automatically converted to subtraction
- ○ **D.** The register expands to 16 bits to accommodate the result

Q19 (2 marks)

In Von Neumann architecture, a program can modify its own instructions during execution.
Why is this possible?

- ○ **A.** Because programs have special privileges that data doesn't have
- ○ **B.** Because the CPU has a special self-modification unit
- ○ **C.** Because instructions and data share the same memory and are stored in the same format, so instructions can be written to like any other data

○ **D.** This is not actually possible in Von Neumann architecture

Q20 (2 marks)

A computer has more running processes than CPU cores.
How does the system handle this situation?

○ **A.** Extra processes are deleted until the number matches the core count

○ **B.** Extra cores are created virtually to match the process count

○ **C.** The computer crashes because it cannot run more processes than cores

○ **D.** The operating system rapidly switches processes between cores, giving each a time slice so all appear to run simultaneously

Q21 (2 marks)

When the CPU requests data from memory location 1000, the cache often also loads data from nearby locations like 1001, 1002, and 1003. Why does the cache do this?

○ **A.** To waste memory by storing unnecessary data

○ **B.** Because spatial locality suggests nearby data will likely be needed soon, so preloading it reduces future cache misses

○ **C.** Because the cache cannot store single memory locations

○ **D.** To slow down the system by loading extra data

Q22 (2 marks)

Two different CPUs have identical clock speeds but different architectures.
Why might they have different performance levels?

- ○ **A.** Clock speed is the only factor that affects performance, so they must perform identically
- ○ **B.** The colour of the CPU package affects performance
- ○ **C.** Different architectures may complete different amounts of work per clock cycle, affecting instructions per second even at the same clock rate
- ○ **D.** Different brand names guarantee different performance levels

Q23 (2 marks)

Computer memory is organised in a hierarchy from registers to secondary storage.
Which statement correctly describes the trade-offs in this hierarchy?

- ○ **A.** All levels have the same speed and capacity
- ○ **B.** Higher capacity always means faster access
- ○ **C.** Registers have the largest capacity because they are inside the CPU
- ○ **D.** Moving down the hierarchy (toward secondary storage), capacity increases while speed and cost per byte decrease

Q24 (3 marks)

When a hardware device needs attention (like a keyboard key press), it sends an interrupt signal to the CPU.
Analyse how the CPU handles this interrupt.

- ○ **A.** The CPU completes its current instruction, saves the current state (including PC value), handles the interrupt, then restores the state to resume the original program
- ○ **B.** The CPU ignores all interrupts until the current program finishes completely
- ○ **C.** The CPU immediately destroys the current program and switches to handling the interrupt permanently
- ○ **D.** Interrupts are handled by the hard drive, not the CPU

Q25 (3 marks)

When the operating system switches from running one program to another, it performs a 'context switch'.
Analyse the CPU operations involved and their performance impact.

- ○ **A.** Context switches are instant and have no performance cost
- ○ **B.** Context switches only involve changing the Program Counter
- ○ **C.** Context switches require saving all CPU registers and state of the current program, loading the saved state of the next program, and updating memory mappings — this takes time and reduces overall throughput
- ○ **D.** Context switches delete the current program from memory

Q26 (3 marks)

A laptop manufacturer must design a CPU that balances performance with battery life and heat generation.
Analyse how clock speed and core count decisions affect these factors.

- ○ **A.** Higher clock speeds and more cores always improve battery life with no downsides

○ **B.** Lower clock speeds reduce power consumption and heat but limit single-threaded performance; more cores increase parallel performance but draw more power — the optimal design balances these trade-offs for mobile use cases

○ **C.** Clock speed and core count have no effect on power consumption or heat

○ **D.** Laptops should always use the fastest available desktop CPU for best results

Q27 (3 marks)

Modern CPUs use 'branch prediction' to guess which way conditional branches (IF statements) will go before the condition is evaluated. Analyse why this technique improves performance and what happens when predictions are wrong.

○ **A.** Branch prediction has no effect on performance because the CPU always waits for condition evaluation

○ **B.** Branch prediction eliminates the need for conditional statements in programs

○ **C.** Branch prediction only works on programs written in a specific programming language

○ **D.** Branch prediction allows the CPU to speculatively fetch and begin executing instructions along the predicted path, keeping the pipeline full; wrong predictions require discarding speculative work and restarting, causing a performance penalty

Q28 (3 marks)

A '64-bit CPU' is described as having a 64-bit word size.
Analyse what this specification affects in the computer system.

- ○ **A.** It determines the size of data the CPU can process in a single operation, the maximum memory addressable, and the range of values registers can hold — affecting overall system capability
- ○ **B.** It only affects the physical size of the CPU chip
- ○ **C.** It means the CPU can only run 64 programs at once
- ○ **D.** It refers to the number of transistors in the CPU

Q29 (3 marks)

A school server runs slowly when many students access it simultaneously. The server has a powerful CPU that shows only 20% utilisation.
Analyse why upgrading to an even faster CPU would likely NOT solve this problem.

- ○ **A.** Faster CPUs always solve all performance problems
- ○ **B.** 20% utilisation means the CPU is working at maximum capacity
- ○ **C.** The CPU is already underutilised at 20%, indicating the bottleneck is elsewhere — likely network bandwidth, storage I/O, or memory — so a faster CPU would sit even more idle while waiting for these slower components
- ○ **D.** Server CPUs cannot be upgraded

Q30 (3 marks)

Modern CPUs include hardware support for virtualisation, allowing one physical computer to run multiple virtual machines.
Analyse how CPU virtualisation support improves performance compared to software-only virtualisation.

- ○ **A.** Hardware virtualisation support has no performance benefit

over software-only approaches

- ○ **B.** Hardware virtualisation eliminates the need for CPUs entirely
- ○ **C.** Virtualisation only works on single-core processors
- ○ **D.** Hardware virtualisation provides CPU instructions that directly support virtual machine operations — allowing efficient execution of guest operating systems without expensive software translation of every instruction, significantly reducing overhead

Total: **55** marks • Time: **30 min**

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